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Title

The effects of student heterogeneity in classrooms on psycho-social outcomes: A systematic review

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Abstract

The current research is a comprehensive systematic review of the effects of heterogeneity of student characteristics within classrooms on psycho-social outcomes. Numerous theories from diverse research fields predict positive and negative effects of within classroom heterogeneity in different dimensions on different psycho-social outcomes. Many of these theories do not state which exact dimensions of heterogeneity are relevant and are also applicable to a range of outcome variables. We therefore decided to conduct a broad review to collect evidence from diverse fields and integrate existing theoretical frameworks. A systematic literature search in three data bases from education, psychology and social science resulted in 59 studies which were included in the review. The results confirmed that the research field is fragmented, yet there are many commonalities. However, results are inconsistent and no clear pattern emerges. We suggest future research to include more different dimensions of heterogeneity, identify mechanisms leading to a relationship between heterogeneity and outcomes and study additional moderators to capture the factors producing variation in this relationship.

Extended Summary

The classroom ecosystem constitutes a crucial context for the psycho-social development of children and adolescents (Bronfenbrenner, 1994). Classroom composition research has acknowledged the considerable impact peers have on students' academic, psycho-social and socio-emotional outcomes (Hattie, 2002; Müller & Zurbriggen, 2016). While there is a long tradition of academic and performance-related classroom composition research (Dar & Resh, 1986; Marsh et al., 2007), research on psycho-social outcomes remains fragmented (Müller & Zurbriggen, 2016). Various disciplines, including education (school effectiveness research), developmental psychology (socialization in the peer group), social psychology and sociology (intergroup processes), political science (research on context effects on trust), criminology (on delinquency) and epidemiology (on mental health) have examined group heterogeneity and its impact on social and psychological outcomes. Theory development and empirical findings have accumulated within disciplinary silos. However, integration of knowledge from these disparate fields could advance our understanding of the ways in which students are affected by their social classroom context.

Classroom composition research generally investigates the influence of aggregated student characteristics such as average academic achievement, average socio-economic background or percentage of ethnic minority students (Dumont et al., 2013) on student outcomes. Heterogeneity has received less attention in educational research, but is often seen as a challenge, for example, because it poses an additional burden on teachers to address diverse needs and potentially problematic social dynamics (Farmer et al., 2019). However, diverse classrooms could also benefit students by exposing them to diverse experiences and perspectives (Piaget, 1932). To date, no comprehensive review about the effects of within-classroom heterogeneity on psycho-social outcomes exists, a gap which the current research addresses.

Our systematic literature search spanned three data bases from education (ERIC), psychology (PsychInfo) and social science (Scopus). The search string consisted of (a) synonyms referring to heterogeneity, (b) to psycho-social outcomes and (c) the context of school, yielding 6,999 results,

with 59 articles included after screening. Many studies excluded from this review did not directly assess heterogeneity, but rather class averages of various characteristics.

Most studies reviewed focused ethnic or linguistic heterogeneity with fewer exploring socio-economic, performance heterogeneity, gender balance, or special educational needs (SEN). Additionally, a wide range of outcomes—socio-emotional traits, social competencies, social behavior, and social relationships—were investigated. Developmental psychology dominated this research (N = 24, followed by sociology (N = 10), education, educational psychology and political science (each N = 5). Other contributions were from economics, social psychology, public health and community psychology, behavior genetics, and criminology. Studies cited diverse theories from social constructivism (Piaget, 1932; Vygotsky, 1978), educational psychology (Olweus, 1994), sociology (Allport, 1954; Blalock, 1967; Blau, 1977; Festinger, 1954), social psychology (Tajfel & Turner, 2004; Turner et al., 1979; Wright, 1986), political science (Putnam, 2007) and criminology (Shaw & McKay, 1942).

Empirical findings are inconclusive: about one-third of studies on ethnic heterogeneity report positive effects, one-third show negative effects, and the rest show no impact. Other dimensions of heterogeneity predominantly yielded negative or null effects, with the exception of heterogeneity in SEN which found positive effects on social relationships. Further subdividing results by outcome domains (cognitive, affective, behavioral) did not reveal different patterns. Following up on our analyses, we will further evaluate evidence for different moderators of the effects of heterogeneity on psycho-social outcomes.

Future studies should broaden the range of heterogeneity dimensions beyond ethnicity and investigate additional moderators to enrich our understanding about the mechanisms by which effects of heterogeneity are transmitted.

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